

Deep Learning

Language Models

Course Contents

Deep Learning

Transformer

LLMs

Context Engineering

Finetuning

Agents

The Bigger Picture

This Part: Deep Learning

- NLP introduction
- neural networks (example: image classification)
- network training (with PyTorch)

Natural Language Processing (NLP)

field within AI and linguistics

aim: enable machines to understand and generate human language

language models: subfield of NLP

aim: model and understand language patterns (typically by learning to predict the next word in a sentence)

capabilities of large language models (LLM) cover most of NLP

NLP Tasks

- part-of-speech tagging
- named entity recognition
- text classification
- machine translation
- summarization
- question answering
- text generation (chatbots)
- code generation
- ...

Symbolic vs Neural NLP

symbolic NLP = rules and logic

- uses hand-crafted rules, lexicons, and grammars
- relies on linguistic knowledge and logic-based systems

neural NLP = data and learning

- uses machine learning (especially deep learning)
- learns patterns from large datasets without explicit rules

Learning from Data



ImageNet

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	NaN	NaN	0	2	2008	WD	Normal	208500
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	NaN	NaN	0	5	2007	WD	Normal	181500
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lv1	AllPub	...	0	NaN	NaN	NaN	0	9	2008	WD	Normal	223500
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lv1	AllPub	...	0	NaN	NaN	NaN	0	2	2006	WD	Abnorml	140000
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lv1	AllPub	...	0	NaN	NaN	NaN	0	12	2008	WD	Normal	250000
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1455	1456	60	RL	62.0	7917	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	NaN	NaN	0	8	2007	WD	Normal	175000
1456	1457	20	RL	85.0	13175	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	MnPrv	NaN	0	2	2010	WD	Normal	210000
1457	1458	70	RL	66.0	9042	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	GdPrv	Shed	2500	5	2010	WD	Normal	266500
1458	1459	20	RL	68.0	9717	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	NaN	NaN	0	4	2010	WD	Normal	142125
1459	1460	20	RL	75.0	9937	Pave	NaN	Reg	Lv1	AllPub	...	0	NaN	NaN	NaN	0	6	2008	WD	Normal	147500

[1460 rows x 81 columns]

The Lord of the Rings

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from Lord of the rings)

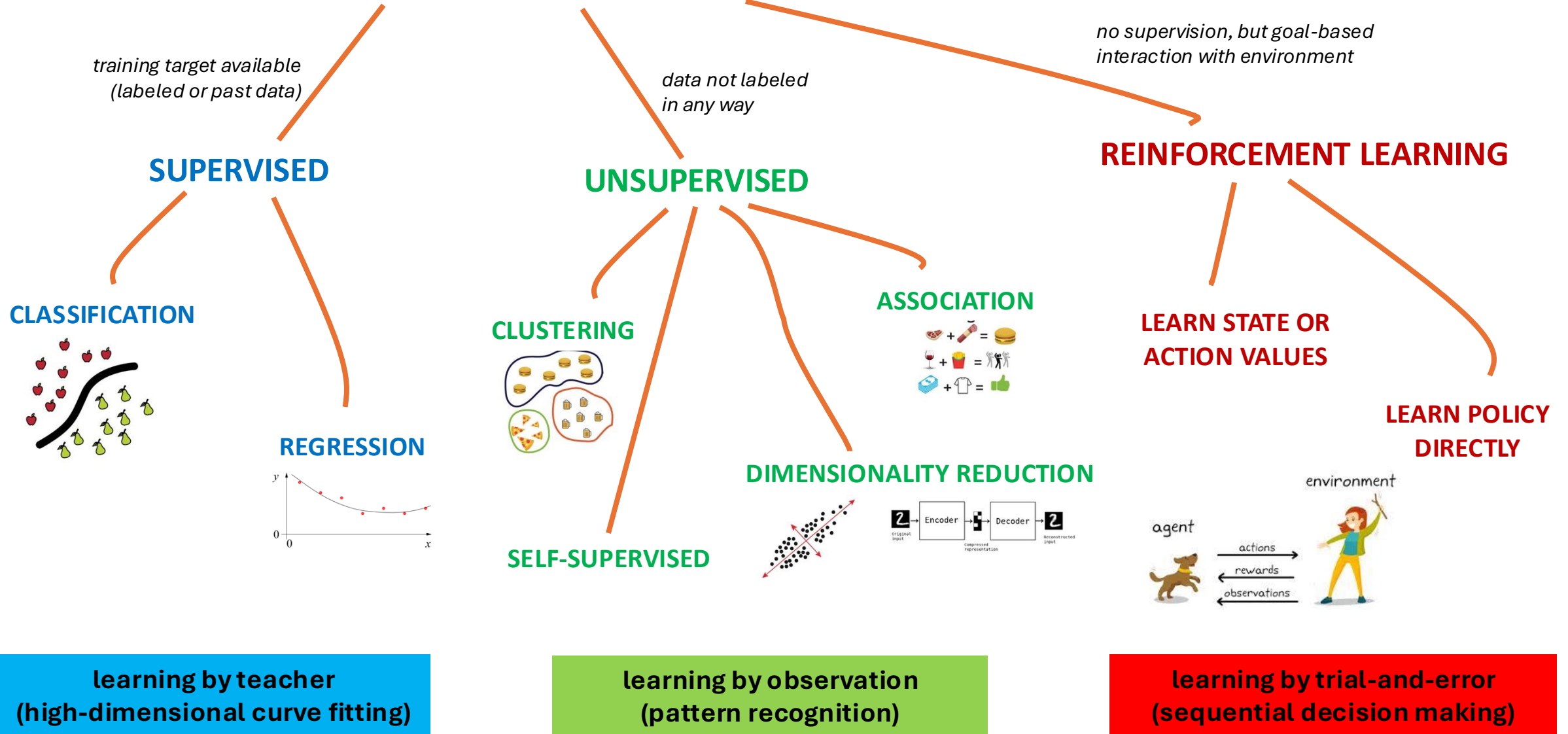
This article is about the book. For other uses, see The Lord of the Rings (disambiguation).
"*War of the Ring*" redirects here. For other uses, see *War of the Ring (disambiguation)*.

The Lord of the Rings is an epic^[1] high fantasy novel^[a] by the English author and scholar J. R. R. Tolkien. Set in Middle-earth, the story began as a sequel to Tolkien's 1937 children's book *The Hobbit*, but eventually developed into a much larger work. Written in stages between 1937 and 1949, *The Lord of the Rings* is one of the best-selling books ever written, with over 150 million copies sold.^[2]

The title refers to the story's main antagonist,^[b] Sauron, the Dark Lord who in an earlier age created the One Ring to rule the other Rings of Power given to Men, Dwarves, and Elves, in his campaign to conquer all of Middle-earth. From homely beginnings in the Shire, a hobbit land reminiscent of the English countryside, the story ranges across Middle-earth, following the quest to destroy the One Ring, seen mainly through the eyes of the hobbits Frodo, Sam, Merry, and Pippin. Aiding Frodo are the Wizard Gandalf, the Men Aragorn and Boromir, the Elf Legolas, and the Dwarf Gimli, who unite in order to rally the Free Peoples of Middle-earth against Sauron's armies and give Frodo a chance to destroy the One Ring in the fire of Mount Doom.

Although often mistakenly called a trilogy, the work was intended by Tolkien to be one volume in a two-volume set along with *The Silmarillion*.^{[3][T 3]} For economic reasons, *The Lord of the Rings* was first published over the course of a year from 29 July 1954 to 20 October 1955 in three volumes rather than one^{[3][4]} under the titles *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*; *The Silmarillion* appeared only after the author's death. The work is divided internally into six books, two per volume, with several appendices of background material.^[c] These three volumes were later published as a boxed set, and even finally as a single volume, following the author's original intent.

MACHINE LEARNING



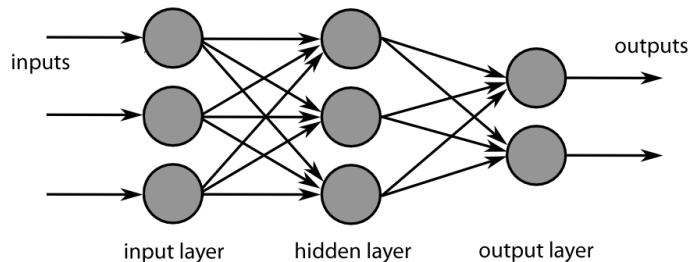
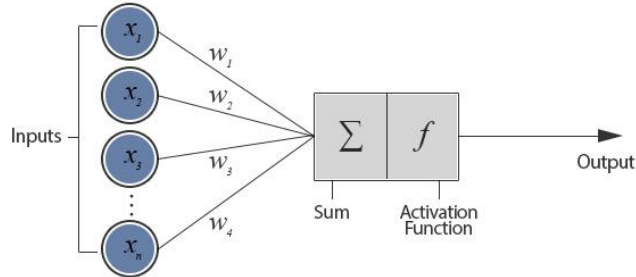
unsupervised and reinforcement learning can
both be cast as supervised-learning setup

Algorithmic Families of Supervised Learning

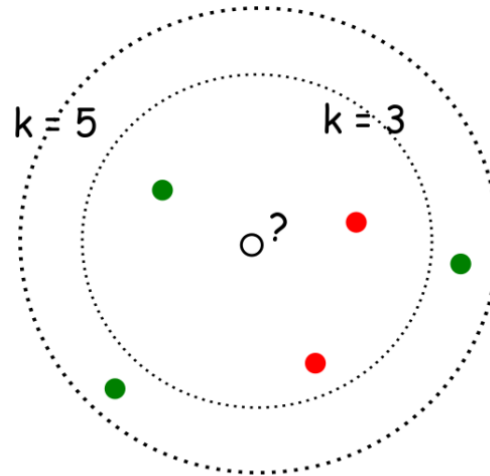
parametric models

linear regression

neural networks: many linear models, nonlinear by means of activation functions



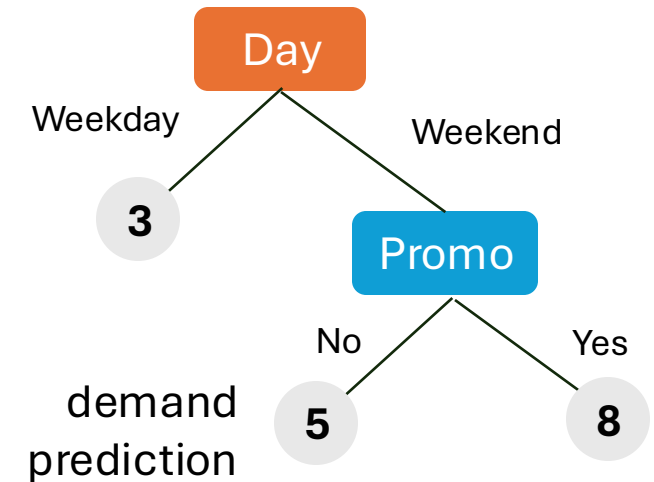
nearest neighbors (local methods, instance-based learning): non-parametric models



with $k = 3$, ●
with $k = 5$, ●

kernel/support-vector machines: linear model (maximum-margin hyperplane) with kernel trick

decision trees: rule learning

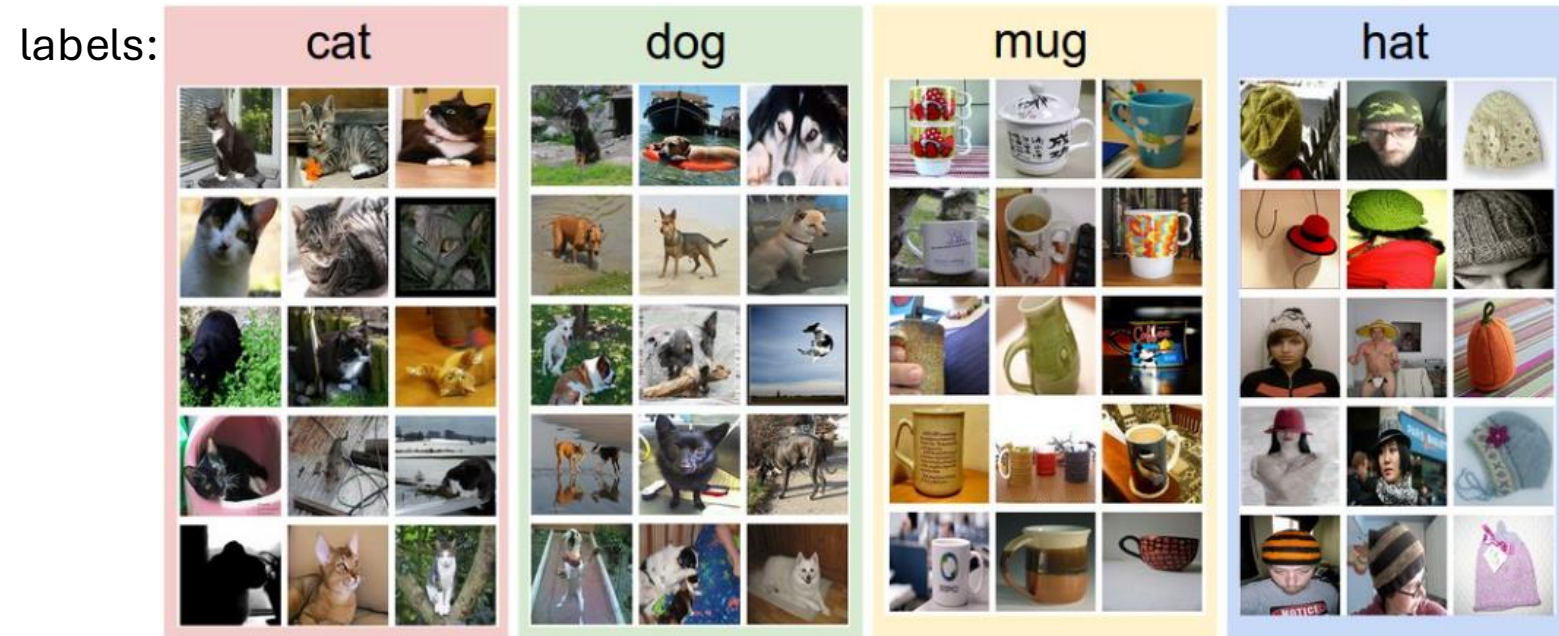


often used in ensemble methods

- bagging: random forests
- boosting: gradient boosting

Example: Image Classification

training data:



test data with learned model:

$$f\left(\text{img of a cat}\right) = \text{"Cat"}$$

$$f\left(\text{img of a dog}\right) = \text{"Dog"}$$

Image Classification as Input-Output Mapping

$$f\left(\text{Image of a Cat}\right) = \text{"Cat"}$$

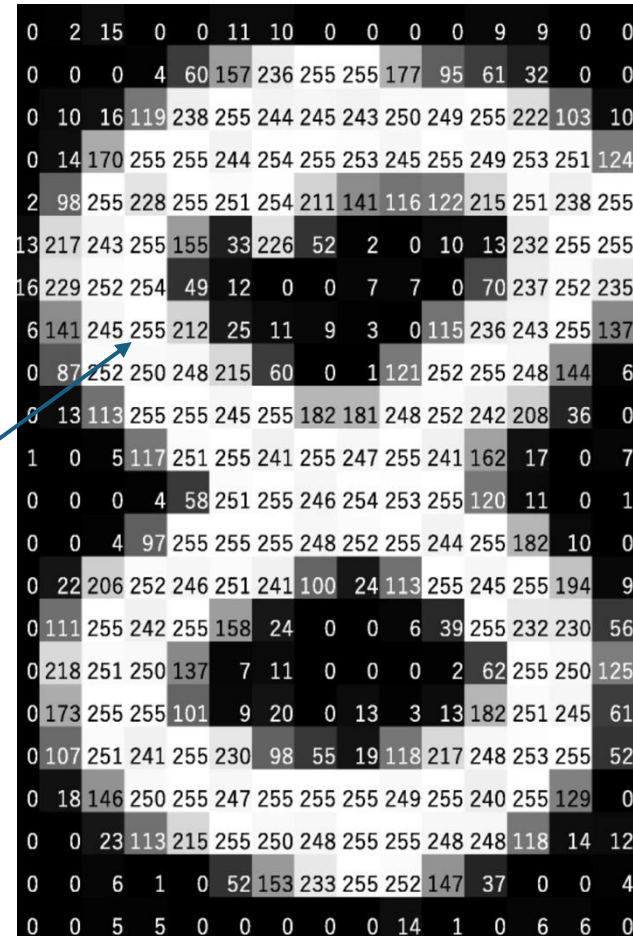
complicated function
→ neural network

$$f\left(\text{Image of a Dog}\right) = \text{"Dog"}$$

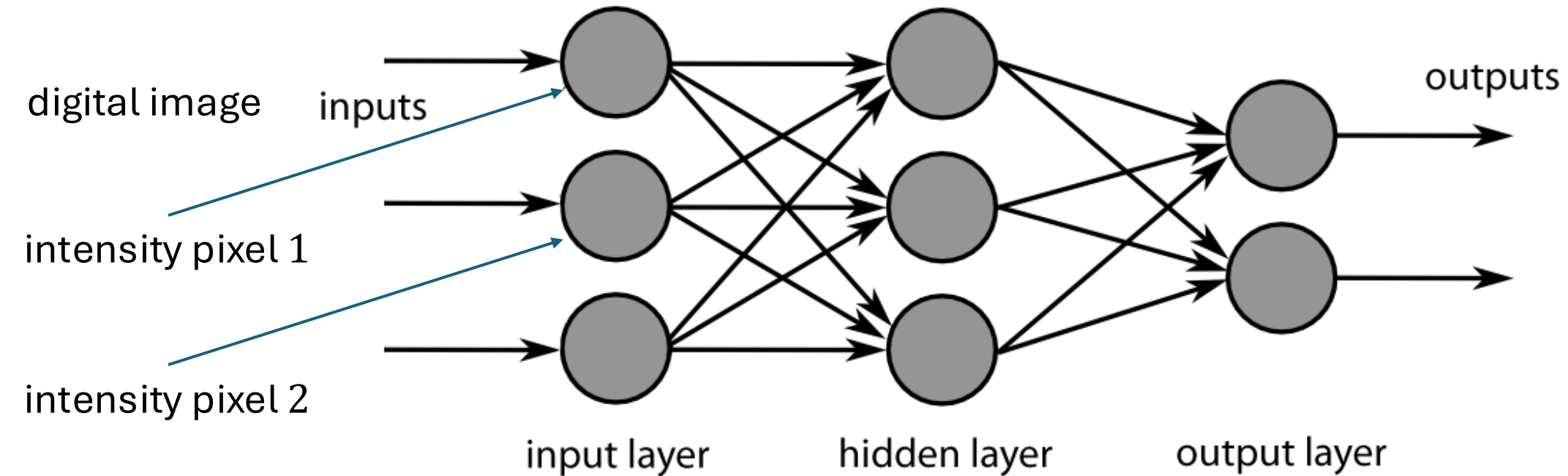
How Computers See Images

digital images:
pixels und intensity

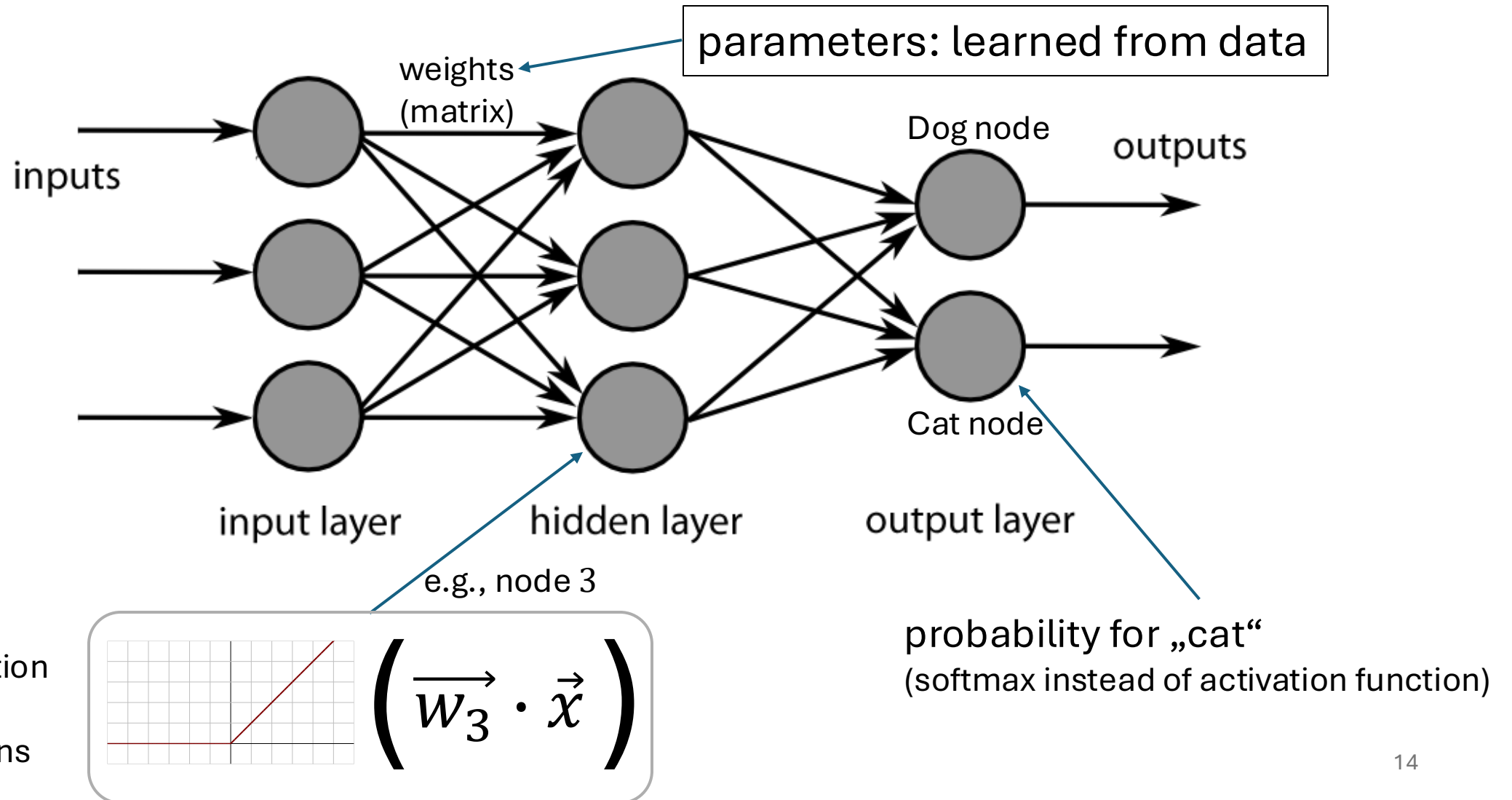
8-bit image



Neural Network as Nonlinear Function

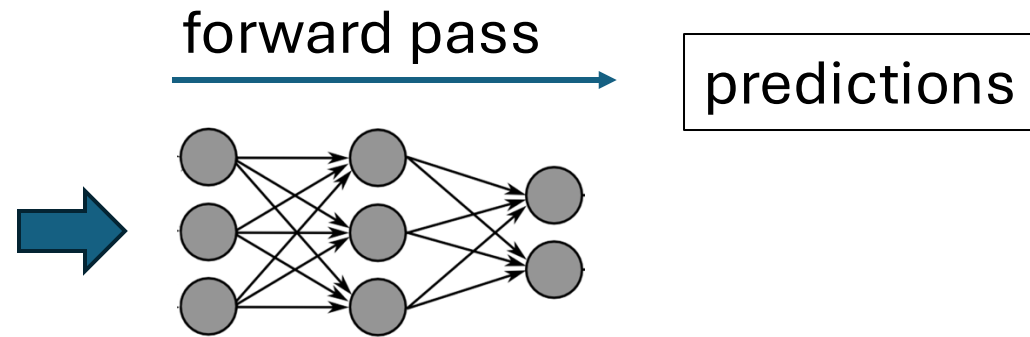
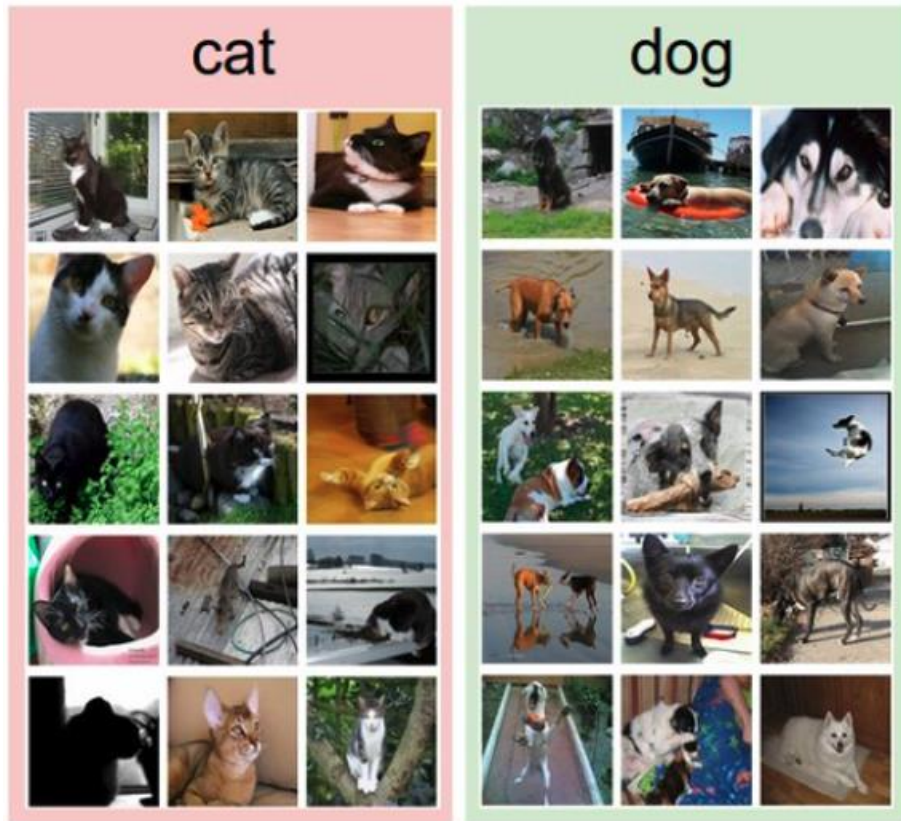


Neural Network as Nonlinear Function



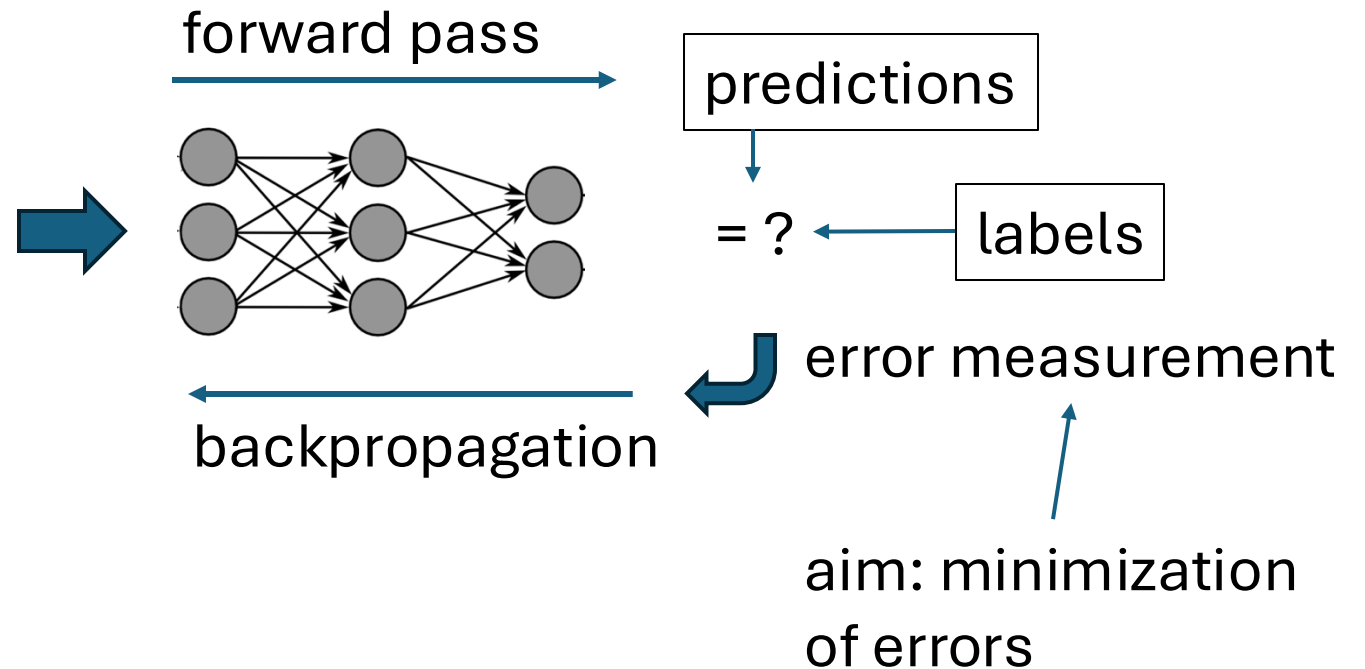
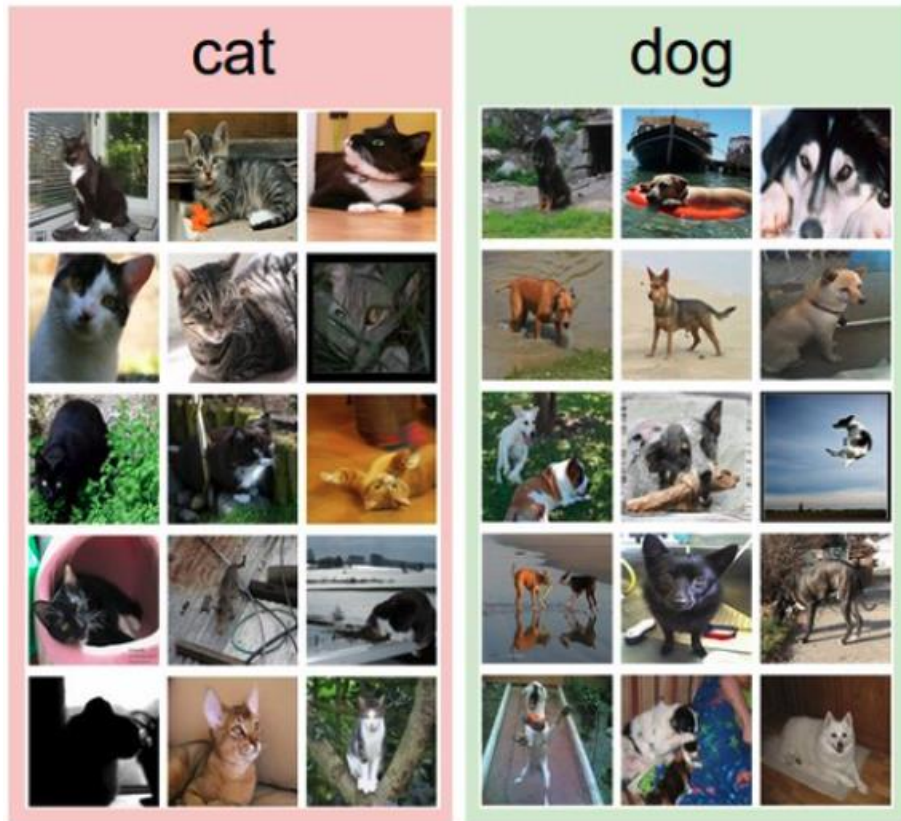
Training with Labeled Images

label:



Training with Labeled Images

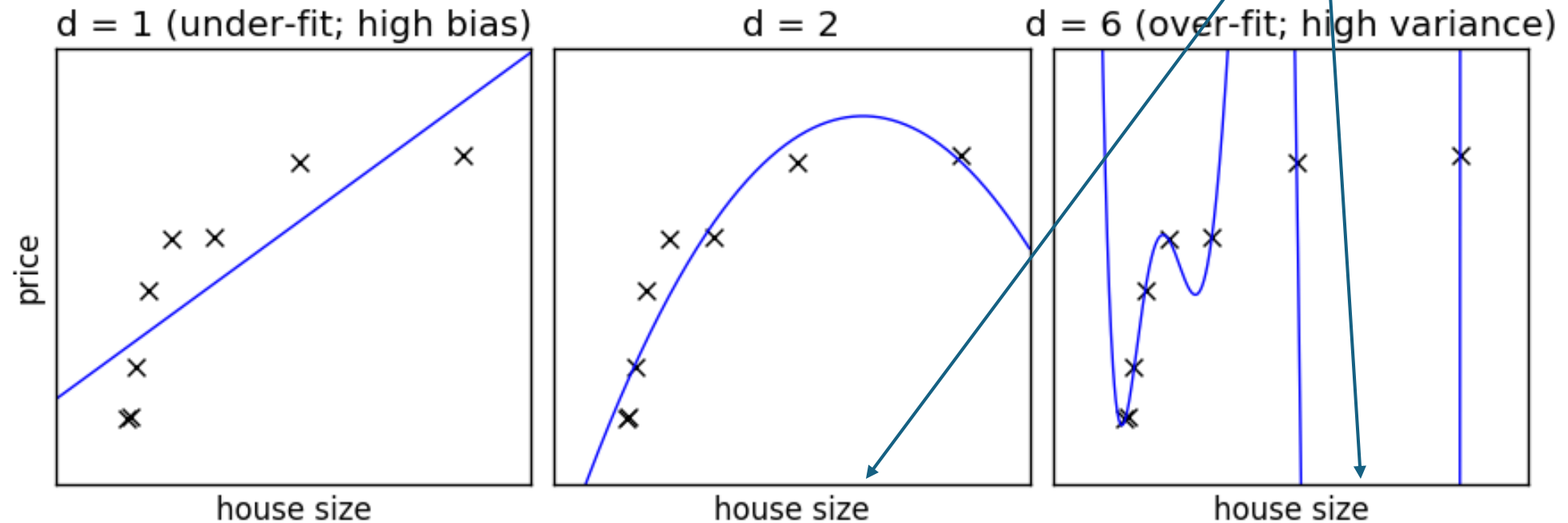
label:



The Training Problem: Under- and Over-Fitting

example: polynomial fit

degree of fitted polynomial



Training: Iterative Adjustment of Weights

```
for epoch in range(epochs):  
    for X, y in mini_batches:  
        pred = model(X)  
        cost = loss(pred, y)  
        cost.backward()  
        optimizer.step()
```

quantification of error
(here: cross entropy)

chain rule to calculate
gradient (direction) of
cost function with
respect to weights

one step in direction of steepest
descent: (stochastic) gradient descent

Classification Networks

cross-entropy loss (one output node for each class k):

$$L(y, \hat{y}) = - \sum_{k=1}^K y_k \log \hat{y}_k$$

using softmax on output nodes:

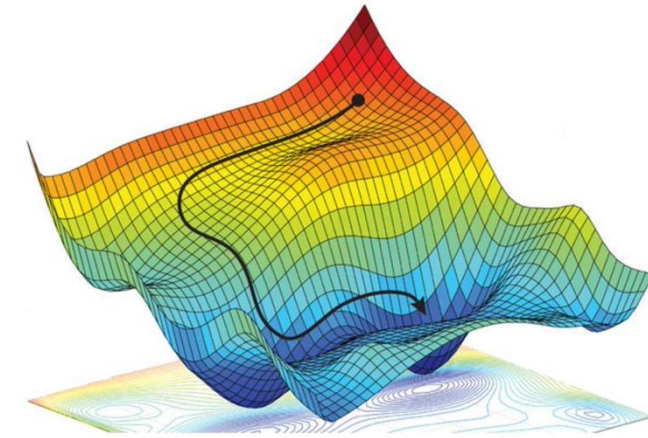
$$\hat{y}_k = \frac{e^{o_k}}{\sum_{l=1}^K e^{o_l}}$$

→ prediction of probabilities

regression networks:

- squared error loss:
 $L(y, \hat{y}) = (y - \hat{y})^2$
- just one output node
- no function on output
→ prediction of real numbers

Stochastic Gradient Descent (SGD)



step size cost function

weight updates: $\hat{\mathbf{w}} \leftarrow \hat{\mathbf{w}} - \eta \nabla_{\hat{\mathbf{w}}} J(\hat{\mathbf{w}})$

options: all training examples

- $J(\hat{\mathbf{w}}) = \frac{1}{n} \sum_{i=1}^n J_i(\hat{\mathbf{w}})$ batch gradient descent (whole training dataset)
- $J(\hat{\mathbf{w}}) = J_i(\hat{\mathbf{w}})$ stochastic gradient descent (single example)
- $J(\hat{\mathbf{w}}) = \frac{1}{m} \sum_{i=1}^m J_i(\hat{\mathbf{w}})$ mini-batch stochastic gradient descent

mini-batch size $\rightarrow$ hyperparameter: tradeoffs between runtime & memory, convergence & generalization

random fluctuations of SGD help to escape saddle points

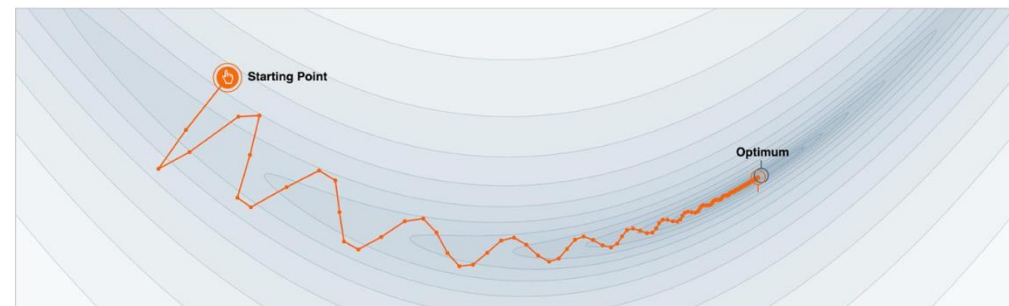
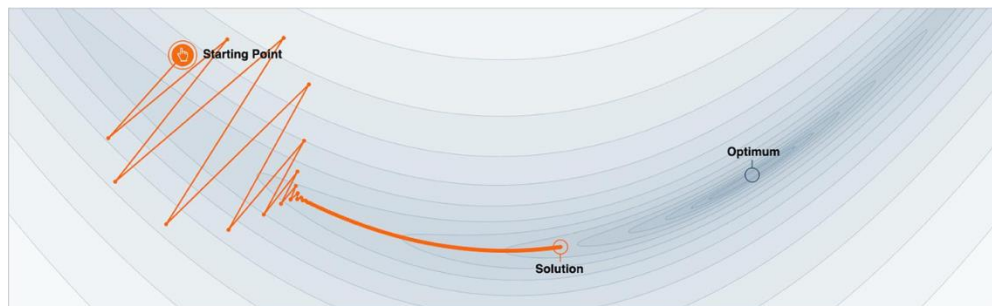
Gradient Descent with Momentum

algorithm:

- estimate gradient: $\mathbf{g} = \nabla_{\hat{\mathbf{w}}} J(\hat{\mathbf{w}})$
- compute velocity update: $\mathbf{v} \leftarrow \alpha \mathbf{v} - \eta \mathbf{g}$
- apply parameter update: $\hat{\boldsymbol{\theta}} \leftarrow \hat{\boldsymbol{\theta}} + \mathbf{v}$

hyperparameter ($0 < \alpha < 1$)
specifying exponential decay

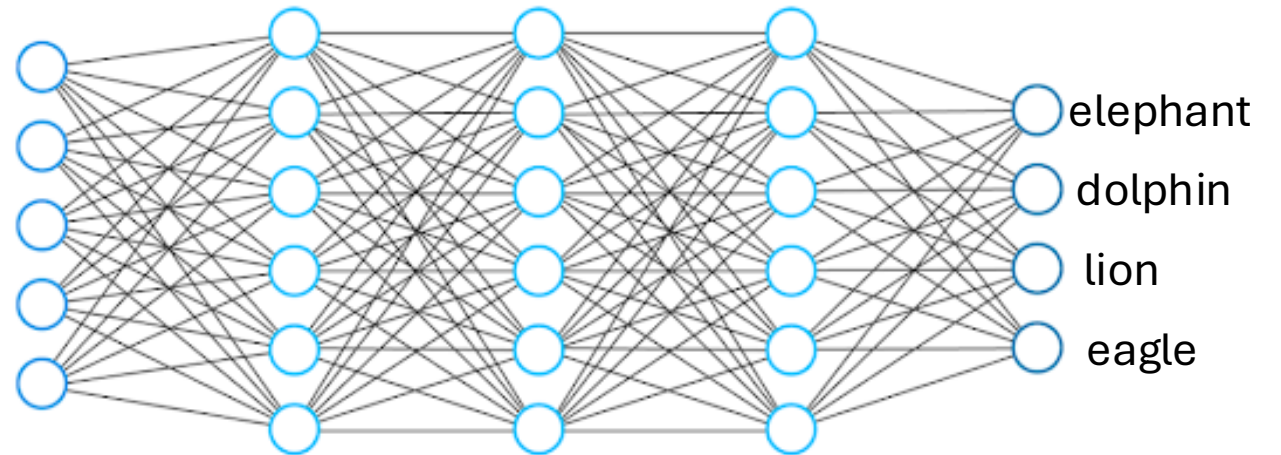
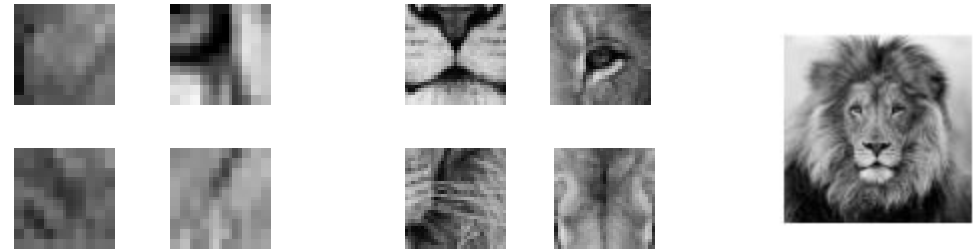
→ no bouncing around of parameters, escape from local minima and saddle points



Hierarchical Representation

deep learning:
several hidden layers

increasing abstraction
→



Side Note: Inductive Bias for Vision

need for consideration of spatial structure

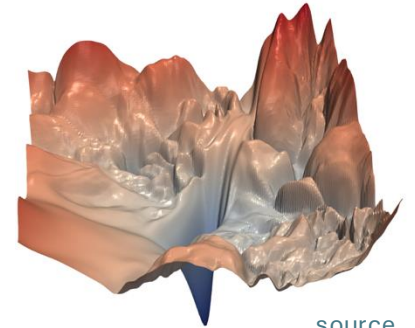
- convolutional neural networks
- vision transformer

How to Train Deep Neural Networks Effectively?

optimization and regularization difficult

- local vs global minima, saddle points, easily overfitting
- many hyperparameters to tune

typical loss surface:



[source](#)

but many methods to get it working in practice (despite partly patchy theoretical understanding)

- optimization: activation and loss functions, weight initialization, adaptive learning rate (e.g., Adam), learning rate schedulers
- explicit regularization: weight decay, dropout, data augmentation
- implicit regularization: early stopping, batch/layer normalization, SGD

Next: Transformer

recent hype around LLMs

fully started with ChatGPT release end of 2022

technical backbone: transformer architecture

transformers also applicable to other areas, for example computer vision (as alternative to convolutional neural networks)